

1. Program to Calculate the Area of a circle and triangle

```
pi=3.14
radius=float(input("PLEASE ENTER RADIUS: ")) #CIRCLE AREA
area=pi*radius*radius
print("Area of circle is: ", area)
```

```
base=float(input("ENTER BASE: ")) #TRIANGLE AREA
height= float(input("ENTER HEIGHT: "))
area=1/2*(base*height)
print(area)
```

2. Program to Swap Two Variables

```
a = input("Enter first variable: ")  
b = input("Enter second variable: ")  
temp=a  
a=b  
b=temp  
print("Swapped values:", a, b)
```

3. Program to generate a random number

importing the random module
import random

print(random.randint(0,100))

4. Program to Convert Kilometers to Miles

```
mile=0.62  
kilometers=float(input("ENTER KM: "))  
print("OUTPUT IS: ", mile*kilometers)
```

5. Program to find maximum of two numbers

```
n1=float(input("ENTER NO1: "))  
n2=float(input("ENTER NO2: "))  
print(max(n1,n2))
```

6. Program to check if a number is even or odd

```
number=float(input("THE NO IS: "))  
if(number%2==0): print("NO IS EVEN")  
else: print("NO IS ODD")
```

7. Program to check if a number is positive, negative or 0

```
n1=float(input("Enter a number: "))
if(n1>0):
    print(n1, "is positive")
elif (n1<0):
    print(n1, "is negative")
else:
    print(n1, "is zero")
```

8. CLASS DESK PROBLEM

```
a = int(input("Enter the students in classroom A: "))
b = int(input("Enter the students in classroom B: "))
c = int(input("Enter the students in classroom C: "))
total_desks = (a + 1) // 2 + (b + 1) // 2 + (c + 1) // 2
print(total_desks)
```


9. Clock problem

```
# Hour hand= 360/12=30 degree  
# Minute hand=30/60=0.5 degree  
# Second hand=0.5/60=1/120degree
```

```
H=int(input("Enter the hour hand value"))  
M=int(input("Enter the minute hand value"))  
S=int(input("Enter the second hand value"))
```

```
angle = (H % 12) * 30 + M * 0.5 + S * (1 / 120)
```

```
print(angle," Degrees")
```

10. Program to Find Fourth Vertex of a Rectangle

```
x1 = int(input("Enter x1: "))
y1 = int(input("Enter y1: "))
x2 = int(input("Enter x2: "))
y2 = int(input("Enter y2: "))
x3 = int(input("Enter x3: "))
y3 = int(input("Enter y3: "))
x4 = x1 if x1 != x2 else x3
y4 = y1 if y1 != y2 else y3
print("Fourth vertex:", x4, y4)
```

11. Program to Find Missing Card Number

```
N = int(input("Enter total number of cards: "))
actual_sum = 0
for i in range(N - 1):
    num = int(input("Enter a remaining card number: "))
    actual_sum += num
expected_sum = N * (N + 1) // 2
print("Missing card number:", expected_sum - actual_sum)
```

12. to find maximum of 3 numbers

```
no1=float(input())  
no2=float(input())  
no3=float(input())  
print(max(no1,no2,no3))
```

13. Program to check if a year is leap

```
y=int(input("Enter the year: "))  
if(y%4==0 and (y%100!=0 or y%400==0)):  
    print(y, " is a leap year")  
else:  
    print(y, " is not a leap year")
```

14. Program to check if a date is valid

```
day = int(input("Enter the dd: "))
month = int(input("Enter the mm: "))
year = int(input("Enter the yyyy: "))

# Basic month days list (non-leap year)
days_in_month = [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31]

# Check if the month is valid
if month < 1 or month > 12:
    print(day, "/", month, "/", year, "is not a valid date.")
else:
    # Adjust February for leap year
    if (year % 4 == 0 and (year % 100 != 0 or year % 400 == 0)):
        days_in_month[1] = 29

# Check if the day is valid
if day <= days_in_month[month - 1]:
    print(day, "/", month, "/", year, "is a valid date.")
else:
    print(day, "/", month, "/", year, "is not a valid date.")
```

15. Program to Find Roots of a Quadratic Equation

```
a = int(input("Enter coefficient a: "))
b = int(input("Enter coefficient b: "))
c = int(input("Enter coefficient c: "))
d = b**2 - 4*a*c
if d < 0:
    print("No real roots")
elif d == 0:
    print("One root:", -b / (2 * a))
else:
    print("Two roots:", (-b + d ** 0.5) / (2 * a), "and", (-b - d ** 0.5) / (2 * a))
```

16. Remove Characters with Indices Divisible by 3

```
s = input("Enter a string: ")
result = ""
for i in range(len(s)):
    if i % 3 != 0:
        result += s[i]
print("Modified string:", result)
```


17. Find Length of Longest Equal Fragment

```
length = 1
max_length = 1
prev = int(input("Enter a number: "))
while True:
    num = int(input("Enter a number: "))
    if num == 0:
        break
    if num == prev:
        length += 1
    else:
        max_length = max(max_length, length)
        length = 1
    prev = num
print("Longest fragment length:", max(max_length, length))
```

18. Bowling Pins Knockdown

```
N = int(input("Enter number of pins: "))
K = int(input("Enter number of rolls: "))
pins = ["|"] * N
for _ in range(K):
    start = int(input("Enter start pin: ")) - 1
    stop = int(input("Enter stop pin: "))
    for i in range(start, stop):
        pins[i] = "."
print("".join(pins))
```

19. Calculate Seconds Between Two Timestamps

```
h1 = int(input("Enter first timestamp hours: "))
m1 = int(input("Enter first timestamp minutes: "))
s1 = int(input("Enter first timestamp seconds: "))
h2 = int(input("Enter second timestamp hours: "))
m2 = int(input("Enter second timestamp minutes: "))
s2 = int(input("Enter second timestamp seconds: "))
t1 = h1 * 3600 + m1 * 60 + s1
t2 = h2 * 3600 + m2 * 60 + s2
print("Seconds between timestamps:", t2 - t1)
```

20. Display First N Numbers

```
n = int(input("Enter n: "))  
print("Series is: ")  
for i in range(1, n + 1):  
    print(i)
```

21. Calculate Factorial

```
num = int(input("Enter a number: "))  
fact = 1  
print("Factorial is: ")  
for i in range(1, num + 1):  
    fact *= i  
print(fact)
```

22. Display Numbers in Reverse Order

```
n = int(input())  
while n > 0:  
    print(n)  
    n -= 1
```

23. Program to check if a number is prime

```
n = int(input("\nEnter a number to check if it's prime: "))
if n > 1:
    for i in range(2, n):
        if n % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime")
else:
    print("Not Prime")
```

24. Program to calculate sum and average of first n numbers

```
n = int(input("Enter a number for sum and average: "))
sum_n = (n * (n + 1)) // 2 # Sum of first n numbers formula
average = sum_n / n
print("Sum:", sum_n)
print("Average:", average)
```


25. Program to display first n multiples of a number

```
num = int(input("Enter a number to display its multiples: "))
n = int(input("Enter how many multiples you want to see: "))
print("\nFirst", n, "multiples of", num, ":")
for i in range(1, n + 1):
    print(num * i)
```

26. Program to display first n Fibonacci numbers

```
n = int(input("Enter a number for Fibonacci sequence: "))
a, b = 0, 1
print("\nFirst", n, "Fibonacci numbers:")
for i in range(n):
    print(a)
    a, b = b, a + b
```

27. Program to find the sum of digits of a number

```
n = int(input("Enter a number: "))
sum_digits = 0
for digit in str(n):
    sum_digits += int(digit)
print("Sum of digits:", sum_digits)
```

28. Program to Create and View Elements of a List

```
my_list = [1, 2, 3, 4, 5]  
print("List:", my_list)
```

29. Program to Create and View Elements of a Tuple

```
my_tuple = (1, 2, 3, 4, 5)  
print("Tuple:", my_tuple)
```

30. Program to Access List Index and Values

```
my_list = [1, 2, 3, 4, 5]
for i in range(len(my_list)):
    print("Index", i, "Value", my_list[i])
```

31. Program to Add Two Lists

```
list1 = [1, 2, 3]
```

```
list2 = [4, 5, 6]
```

```
print("Added Lists:", list1 + list2)
```

32. Program to Check if a List is Empty or Not

```
if not my_list:  
    print("List is empty")  
else:  
    print("List is not empty")
```


33. Program to Find the Largest Number in a List

```
my_list = [10, 20, 5, 30]  
print("Largest number:", max(my_list))
```

34. Program to Find the Second Largest Number in a List

```
my_list=[5,30,10,2]  
my_list.sort()  
print("Second largest number:", my_list[-2])
```

35. Program to Put Even and Odd Elements in a List into Two Different Lists

```
my_list=[1,2,3,4,5,6]
even_list = []
odd_list = []
for i in my_list:
    if i % 2 == 0:
        even_list.append(i)
    else:
        odd_list.append(i)
print("Even numbers:", even_list)
print("Odd numbers:", odd_list)
```

36. Program to Find all Numbers in a Range which are Perfect Squares and Sum of all Digits in the Number is Less than 10

```
for num in range(1, 50):
    sqrt = num ** 0.5
    if sqrt == int(sqrt):
        num_str = str(num)
        sum_of_digits = 0
        for digit in num_str:
            sum_of_digits += int(digit)
        if sum_of_digits < 10:
            print("Perfect square:", num)
```

37. Program to Generate Random Numbers from 1 to 20 and Append Them to the List

```
import random
random_numbers = []
for i in range(5):
    num = random.randint(1, 20)
    random_numbers.append(num)
print("Random numbers:", random_numbers)
```

38. Program to Remove the Duplicate Items from a List

```
my_list_with_duplicates = [1, 2, 2, 3, 4, 4]
unique_list = list(set(my_list_with_duplicates))
print("List without duplicates:", unique_list)
```

39. Program to Create and View a Dictionary

```
student = {  
    "Name": "Abhishek",  
    "Age": 8,  
    "Course": "CSE-AIML",  
    "College": "VIT"  
}  
print("Student Dictionary:", student)
```

40. Program to Create and View Elements of a Set

```
my_set = {1, 2, 3, 4}  
print("Set:", my_set)
```


41. Program to Print Values of Dictionary

```
dictionary= {  
    "car":"honda",  
    "color":"red",  
    "engine": "v12"  
}  
print(dictionary.values())
```

42. Program to Print all Keys of Dictionary

```
dictionary= {  
    "car":"honda",  
    "color":"red",  
    "engine": "v12"  
}  
print(dictionary.keys())
```

43. Program to Insert and Delete from Dictionary

```
my_dict = {'a': 1, 'b': 2, 'c': 3}
my_dict['d'] = 4 # Insert
del my_dict['a'] # Delete
print("Updated Dictionary:", my_dict)
```

44. Program to Sort (Ascending and Descending) a Dictionary by Value

```
my_dict = {'a': 1, 'c': 3, 'b': 2,}
asc = dict(sorted(my_dict.items(), key=lambda item: item[1]))
desc = dict(sorted(my_dict.items(), key=lambda item: item[1], reverse=True))
print("Sorted (Ascending) Dictionary:", asc)
print("Sorted (Descending) Dictionary:", desc)
```

45. Program to Concatenate Dictionaries to Create a New One

```
dict1 = {'a': 1, 'b': 2}
dict2 = {'c': 3, 'd': 4}
concatenated_dict = {**dict1, **dict2}
print("Concatenated Dictionary:", concatenated_dict)
```

46. Program to Check Whether a Given Key Already Exists in a Dictionary

```
my_dict = {'a': 3, 'b': 1, 'c': 2}
key = input("Enter the key to check: ")
if key in my_dict: # Check if key exists
    print("Key", key, "exists in the dictionary")
else:
    print("Key", key, "does not exist in the dictionary")
```

47. Program to Merge Two Python Dictionaries

```
dict1 = {'x': 10, 'y': 20}
dict2 = {'z': 30, 'w': 40}
merged_dict = dict1.copy()
merged_dict.update(dict2)
print("Merged Dictionary:", merged_dict)
```

48. Program to Get the Maximum and Minimum Value in a Dictionary

```
my_dict={'a':1, 'b':2, 'c':100, 'd':4}
max_value = max(int(value) for value in my_dict.values())
min_value = min(int(value) for value in my_dict.values())

print("Maximum value in dictionary:", max_value)
print("Minimum value in dictionary:", min_value)
```


49. Program to Add a List of Elements to a Set

```
my_set={1,2,3,4}  
my_set.update([5, 6, 7])  
print("Updated Set:", my_set)
```

50. Program to Update the First Set with Items that Don't Exist in the Second Set

```
set1 = {1, 2, 3}
set2 = {3, 4, 5}
set1.update(set2 - set1)
print("Set1 after update:", set1)
```

51. Elements present in Set A or B, but not both

```
set1 = {1, 2, 3, 4}
```

```
set2 = {3, 4, 5, 6}
```

```
result = set1 ^ set2
```

```
print("Elements present in Set A or B, but not both:", result)
```

52. Check if sets have any common elements

```
set1 = {1, 2, 3, 4}
```

```
set2 = {3, 4, 5, 6}
```

```
if set1 & set2: # Intersection is not empty
```

```
    print("Sets have common elements.")
```

```
else:
```

```
    print("Sets have no common elements.")
```

53. Remove items from set1 that are not common to both set1 and set2

set1 = {1, 2, 3, 4}

set2 = {3, 4, 5, 6}

53. Remove items from set1 that are not common to both set1 and set2

set1 &= set2 # Intersection of set1 and set2

print("Set1 after removing non-common elements:", set1)

54. Function to check if a number is even or odd

```
def check_even_odd(num):  
    if num % 2 == 0:  
        return "Even"  
    else:  
        return "Odd"  
num=int(input("Enter a number: "))  
print(check_even_odd(num))
```

55. Function to find the maximum of two numbers

```
def find_max(num1, num2):  
    return num1 if num1 > num2 else num2  
num1=int(input("Enter the first number: "))  
num2=int(input("Enter the second number: "))  
print("The maximum number is:", find_max(num1, num2))
```

56. Function with keyword arguments

```
def greet_person(name="Guest"):
    return f"Hello, {name}!"

print(greet_person("Abhishek"))
```


57. Function with default arguments

```
def greet_with_default(name="Guest", message="Welcome!"):
    return f"Hello, {name}. {message}"
print(greet_with_default("Abhishek", "You are great!"))
```

58. Function to find the factorial of a number using recursion

```
def factorial(n):  
    if n == 0:  
        return 1  
    else:  
        return n * factorial(n - 1)  
n=(int(input("Enter a number: ")))  
print("Factorial of ",n," is: ",factorial(n))
```

59. Function to find the sum of the digits of the number recursively

```
def sum_of_digits(n):  
    if n == 0:  
        return 0  
    else:  
        return n % 10 + sum_of_digits(n // 10)  
n=int(input("Enter the number:"))  
print("Sum of digits: ",sum_of_digits(n))
```

60. Function to count vowels in a string

```
def count_vowels(string):  
    vowels = "aeiouAEIOU"  
    return sum(1 for char in string if char in vowels)  
string=input("enter the string: ")  
print(count_vowels(string))
```

61. Function greet user that takes a user's name as a parameter and prints a personalized greeting message

```
def greet_user(name):  
    print("Hello", name, "Welcome to Shark Tank.")  
name = input("Enter your name: ")  
greet_user(name)
```

62. Function to print the sequence in reverse order using recursion

```
def reverse_sequence():
    num = int(input("Enter a number: "))
    if num == 0:
        return
    else:
        reverse_sequence() # Recursive call
        print(num,end=" ")

reverse_sequence()
```